Shenzhen Standard Water Technology Limited
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氯气发生器 DVT 阶段老化测试---测试计划

Chlorine Generator DVT Stage Aging Test - Test Plan

测试名称：氯气发生器老化测试

Test Name: Chlorine Generator Aging Test

测试样机：2 号机

Test Prototype: Unit 2

测试地点：深圳

Test location: Shenzhen, China

测试时间：计划累积运行 150 小时以上；

Test time: Plan to accumulate over 150 hours of operation

计划版本：~~V3.0~~ V3.1

SWC 审阅 HSW V3.0，供 10/14 讨论 [10/13 SWC]

Planned version: ~~V3.0~~ V3.1

SWC review of HSW V3.0, for discussion 10/14 [10/13 SWC]

测试人员：赵学平/袁帅

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0、10/13 SWC 修改说明

0. Changes in V3.1 (10/13 SWC)

本版本在 HSW V3.0 的基础上直接修改。所有修改均为红色并标注“10/13 SWC”，删除的文字以红色删除线显示。下表列出每处修改、原因及其对应的 9/28 意见或 SWC 10/13 审阅要点。用量计算见《Aging test materials list 2026-10-13.xlsx》的 Calc 表。

This version edits HSW V3.0 in place. Every change is in red and tagged "10/13 SWC", and deleted words are shown in red strikethrough. The table lists each change, the reason, and the 9/28 comment or 10/13 SWC review point it answers. The quantities are worked out on the Calc sheet of "Aging test materials list 2026-10-13.xlsx".

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# #	修改 **Change**	原因 **Reason**	对应的意见 **Answers**
1	测试目的 2、4、6：目的 2 英文与中文一致；目的 4 耗盐量以称重为准，EC1 作对照；目的 6 注明主测试定时排碱，触发逻辑在 150 小时后验证。 Objectives 2, 4 and 6. The English of objective 2 now matches the Chinese. Objective 4 takes salt use from weighed additions, with EC1 as a cross-check. Objective 6 notes the scheduled drain in the main test and the trigger tests after 150 h.	中英文不一致；单用 EC1 判断耗盐不可靠。 English and Chinese differed. EC1 alone isn't reliable for salt use.	9/28 其他意见；9/28 第 3 条 9/28 other points; 9/28 comment 3
2	测试目的前增加：150 小时能说明什么。 Above the objectives: what 150 h can show.	150 小时可以看到腐蚀起始和早期失效，不能证明水泵寿命。 150 h shows corrosion onset and early failures, not pump life.	V2.3 §0 第 12 条 V2.3 §0 item 12
3	备注 2：请 HSW 确认场地（楼顶或室内）、每天运行时间和夜间状态、开始和结束日期；开始前记录 2 号机配置。 Note 2. HSW to confirm the site (rooftop or indoor lab), daily run hours, the overnight state, and start and end dates. Record Unit 2's configuration before start.	9/28 起未回复；室内时氢气会从水箱逸出，需要通风；配置记录使结果可追溯。 Open since 9/28. Indoors, H2 comes out of the return water in the tank and needs ventilation. The configuration record makes results traceable.	9/28 其他意见；9/28 第 7 条 9/28 other points; 9/28 comment 7
4	备注 a：600 mg/L 按最恶劣工况测试，不是加速老化。 Note (a). 600 mg/L is worst-case testing, not acceleration.	USGS 分档针对自来水；泳池水因蒸发和补水会更硬。 The USGS bands describe tap water. Pool water runs harder through evaporation and refills.	9/28 第 1 条 9/28 comment 1

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# #	修改 **Change**	原因 **Reason**	对应的意见 **Answers**
5	方法 1：每次换水后先补足该阶段的硬度和碱度再开机；列出每阶段投加量；指定硬度检测方法。 Method 1. After every water change, re-dose to the phase's hardness and alkalinity before restarting. Doses per phase listed. Hardness test named.	否则 25 小时后的测试都在较软的水中进行；五合一水质仪不能测硬度，清单中没有硬度检测。 Otherwise every segment after 25 h runs on softer water than its phase calls for. The five-in-one meter doesn't read hardness, and the list has no hardness test.	9/28 第 1、4 条 9/28 comments 1 and 4
6	方法 1：增加泳池 pH 和碱度的维持方法：随电极连续投加碱液（优先），或每 2 小时人工补加碳酸氢钠。 Method 1. How pool pH and alkalinity are held: base dosed continuously with the electrodes (preferred), or NaHCO3 by hand at each 2 h check.	满负荷时每小时约 1.1 mol 酸进入水箱，消耗约 56 mg/L 碱度，初始碱度 2-3 小时用完；pH 下降后碳酸钙垢无法形成。 At full output about 1.1 mol of acid per hour reaches the tank and uses about 56 mg/L of alkalinity, so the starting alkalinity lasts 2-3 h. Once pH drops, CaCO3 scale can't form.	9/28 第 1 条；SWC 10/13 决定 9/28 comment 1; SWC decision 10/13
7	方法 2：确认 DPD 分析仪型号与试剂匹配，试剂量覆盖约 1000 次测试；英文 Dongrun/Bochu 改为 BOQU。 Method 2. Confirm the DPD analyzer model and that its reagents match, enough for about 1,000 tests. English "Dongrun/Bochu" corrected to BOQU.	清单试剂为 SWCL-300，现场分析仪为博取。 The listed reagents are for the SWCL-300, and the analyzer on site is a BOQU.	SWC 10/13 审阅 SWC review 10/13

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# #	修改 **Change**	原因 **Reason**	对应的意见 **Answers**
8	方法 3：主测试中定时手动排碱（D 阀），记录时间、排放量和前后盐水 pH；“腐蚀性水平传感器”改为“碱液液位传感器”。 Method 3. Scheduled manual valve-D drain in the main test, logged with time, volume and brine pH before and after. "Corrosive level sensor" corrected to "caustic level sensor".	至 v1.9.108 固件不会自动打开 D 阀，与 HSW 10/8 的判断一致。不排碱时盐水 pH 会升到 11 的停止线，同时水箱承担全部酸负荷。 Up to v1.9.108 the firmware never opens valve D on its own, which matches HSW's 10/8 note. Without a drain, brine pH climbs to the pH 11 stop while the tank takes the whole acid load.	9/28 第 3 条；SWC 10/13 决定 9/28 comment 3; SWC decision 10/13
9	方法 4：规定余氯测试的固定条件；低于 4.4 mg/L 标记，低于 3.4 mg/L 停止。 Method 4. Fixed conditions for the chlorine check. Flag below 4.4 mg/L, stop below 3.4.	原文只规定了水量和重复次数；4.4 mg/L 为产品指标。 Only the volume and the repeats were set. 4.4 mg/L is the product spec.	9/28 第 4 条 9/28 comment 4
10	方法 5：回答过氧化氢投加量；选用约 1-10 mL/min 变速蠕动泵连续投加，不用定时器；用量约 26-38 kg。 Method 5. Answers HSW's H2O2 dose question. A variable-speed peristaltic pump of about 1-10 mL/min, running continuously with no timer. About 26-38 kg needed.	WS-20-03-M（20 L/h）只用到约 1%额定流量；定时间歇投加会使余氯读数波动。 The WS-20-03-M (20 L/h) would run at about 1% of rating. Timed cycling makes FCL swing in the DPD data.	9/28 第 5 条；HSW 10/8 问题；SWC 10/13 决定 9/28 comment 5; HSW question 10/8; SWC decision 10/13
11	方法 6：增加温度通道；盐水温度探头规格；记录仪统一使用 UTC。 Method 6. More temperature channels, a brine probe spec, and UTC on all logger clocks.	清单中的探头为铝水锌水专用；DaMonTo 使用 UTC。 The listed probe is made for molten aluminium and zinc. DaMonTo stores UTC.	9/28 第 6 条；Arlind 9/30 9/28 comment 6; Arlind 9/30

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# #	修改 **Change**	原因 **Reason**	对应的意见 **Answers**
12	方法 7：“水 pH 值”改为“盐水 pH 值 PH1”；增加 4.4 mg/L 标记线；水样变色以白卡对照拍照。 Method 7. "Water pH" becomes "brine pH, PH1". A 4.4 mg/L flag line. Discoloration judged from photos against a white card.	pH 11 停止线针对盐水，不是泳池水；变色判断需要参照。 The pH 11 stop applies to the brine, not the pool. Discoloration needs a reference.	9/28 第 7 条 9/28 comment 7
13	方法 8：删除“当发现异常情况需终止测试”；更正探测器安装位置；探测器需要输出以便记录；建议报警继电器就地切断电解电源。 Method 8. Removed "the test should be terminated". Corrected detector placement. Detectors need an output so readings are logged. We recommend the alarm relay cuts the electrode supply locally.	该句与暂停检查的规定矛盾；清单中的报警器只能显示和报警。 That clause conflicted with the pause-and-inspect rule. The listed detectors only display and alarm.	9/28 第 8 条 9/28 comment 8
14	方法 9：DaMonTo 为主记录，人工读数用于校核；A、B 两槽分开记录；设备 ID 发给 Incodeks 归档；补充中文。 Method 9. DaMonTo and the loggers are the record, and manual readings check them. Cells A and B logged separately. Device IDs to Incodeks for the archive. Chinese added for the English sentence.	DaMonTo 原始消息只保存 90 天。 Raw DaMonTo messages are deleted after 90 days.	9/28 第 9 条；Arlind 9/30 9/28 comment 9; Arlind 9/30
15	方法 10：拆解时在盐水箱和文丘里 T 型口各取水垢样品，用稀盐酸鉴别。 Method 10. At teardown, scale samples from the brine tank and the venturi T, tested with dilute acid.	区分碳酸钙垢和氢氧化镁垢。 Tells CaCO3 scale from Mg(OH)2.	9/28 第 10 条；V2.3 §6 9/28 comment 10; V2.3 §6

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# #	修改 **Change**	原因 **Reason**	对应的意见 **Answers**
16	记录参数：增加泳池 pH 和总碱度、硬度、排碱、Cl2 和 H2 读数、环境温湿度、补碱记录。 Record parameters. Added pool pH and alkalinity, hardness, caustic drains, Cl2 and H2 readings, ambient temperature and humidity, and base additions.	与方法 1、3、8 一致。 Matches methods 1, 3 and 8.	9/28 第 1、3、8 条 9/28 comments 1, 3 and 8

需在 10/14 确认 / To confirm on 10/14

- 时间基准：记录仪用 UTC；人工记录用北京时间并注明 UTC+8。
  Time base: UTC on loggers; manual sheets in Beijing time, marked UTC+8.
- 余氯测试：低于 4.4 mg/L 标记，低于 3.4 mg/L 停止。
  Chlorine check: flag below 4.4 mg/L, stop below 3.4.
- 排碱：间隔、每次时长和操作方法由 HSW 提出。
  Caustic drain: HSW to propose the interval, duration and method.
- 开始和结束日期：由 HSW 提出。
  Start and end dates: HSW to propose.
- 采购：泵、盐水温度探头、气体探测器输出和数量变更，见《Aging test materials list 2026-10-13.xlsx》。

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Purchasing: the pump, brine probe, detector outputs and quantity changes in "Aging test materials list 2026-10-13.xlsx".

- Incodeks：确认 v1.10.132 是否会自动打开 D 阀，以及运行结束后的冲洗包括哪些动作。

  Incodeks: confirm whether v1.10.132 opens valve D on its own, and what the end-of-run rinse does.

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10/13 SWC：150 小时可以看到腐蚀起始和早期失效，不能证明水泵的额定寿命。

10/13 SWC: 150 h shows corrosion onset and early failures. It does not demonstrate the pump's rated life.

测试目的：

1、验证 DVT 样机长时间运行各部件（增压水泵，压力传感器，流量计，电解电极）老化和性能衰减情况统计；

Test objective: To verify the statistical analysis of aging and performance degradation of various components (booster pump, pressure sensor, flow meter, electrolysis electrode) of DVT samples during long-term operation.

2、观察高硬度水，在持续运行后对流路循环系统的影响（结垢堵塞水路）；

~~Observe the continuous operation of the flow path circulation system;~~Observe how high-hardness water affects the flow path after continuous running (scale blocking the water path); [10/13 SWC]

3、验证 DVT 样机长时间运行（盐水温度，盐水 PH）变化与电解电解电极健康状况的关系；

Verify the relationship between the long-term operation of DVT samples (saltwater temperature, saltwater pH) and the health status of electrolytic electrodes;

4、验证 DVT 样机长时间运行盐的消耗量和水的投加量~~（统计 EC1 数值判断盐消耗量）~~（以称量的加盐量为准，EC1 数值作对照）[10/13 SWC];

Verify the long-term salt consumption and water dosage of DVT samples during operation ~~(determine salt consumption based on EC1 values)~~(salt consumption from the weighed salt additions, with EC1 as a cross-check) [10/13 SWC];

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5、验证未来使用 ORP 数值作为氯投加的参考依据的可靠性（使用 DPD 做对比）；

Verify the reliability of using ORP values as a reference for chlorine dosing in the future (using DPD for comparison);

6、验证排碱逻辑是否可靠，（运行结束后排放--盐水高液位--PH 过高）是否可以有效控制盐水 PH； **10/13 SWC：**主测试按方法 3 定时排碱；PH 触发排碱和碱液液位传感器在 150 小时后分阶段验证。

Verify the reliability of the alkali discharge logic and whether it can effectively control the pH of the salt water (Discharge after operation - high saltwater level - high pH value）； **10/13 SWC:** The main test drains on the schedule in method 3. The pH-triggered drain and the caustic level sensors are verified in the phases after 150 h.

备注：

1、考虑到 EVT 样机在美国的测试中，出现了碱性物质结垢---文丘里负压口堵塞--高硬度（未过滤的水）进入盐水箱---产生的酸性物质进入盐水箱---加速堵塞的循环堵塞；我们在做各部件老化测试中将人为提高泳池循环水中的硬度，从而记录设备长时间运行的流路畅通情况，观察水硬度提高对整个系统的冲击；

Note: Considering that the EVT prototype experienced alkaline scaling during testing in the United States, blockage of the Venturi negative pressure port, high hardness (unfiltered water) entering the brine tank, acidic substances entering the brine tank, and

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accelerated blockage in the cycle; We will artificially increase the hardness of the circulating water in the swimming pool during the aging test of various components, in order to record the smooth flow path of the equipment during long-term operation and observe the impact of the increase in water hardness on the entire system;

2、现有测试环境中我们只能测试一台 DVT 样机 2#和一台 DPD 样机，初始阶段我们将使用纯净的水和盐开启测试；

In the current testing environment, we can only test one DVT prototype 2 # and one DPD prototype. In the initial stage, we will use pure water and salt to initiate the testing;

10/13 SWC：请 HSW 确认：(1) 测试场地在楼顶还是室内实验室；(2) 每天运行时间和夜间状态；(3) 计划开始和结束日期。如在室内，水箱上方需要通风，因为氢气会从回水中逸出。开始前记录 2 号机配置：隔膜型号、电极组编号、文丘里型号、主板和屏幕固件版本、盐的等级和来源。

10/13 SWC: Please confirm (1) whether the rig is on the rooftop or in the indoor lab, (2) the daily run hours and the overnight state, and (3) the planned start and end dates. Indoors, ventilate above the tank, because H2 comes out of the return water there. Before starting, record Unit 2's configuration: membrane type, electrode set ID, venturi type, mainboard and screen firmware versions, and salt grade and source.

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测试方法：

1、使用 1000L 水箱模拟泳池循环，开始状态向设备内部投加纯净的盐和纯净的水，以双通道 15A 电流进行电解，为提高泳池水中的硬度每 1000L 水中加入 162g 无水氯化钙（AR96％）和 60g 无水氯化镁（AR96％），来模拟美国‘极硬水’（以 $CaCO_3$ 计 200 毫克/L），根据三个梯度测试，第一梯度前 50 小时（以 $CaCO_3$ 计 200mg/L），第二梯度中间 50 小时（以 $CaCO_3$ 计 400mg/L）第三梯度最后 50 小时（以 $CaCO_3$ 计 600mg/L），投加方式采用一次性投加中间根据监测数据（手持硬度测试仪）补加的方式，每两小时记录一次硬度读数；添加碳酸氢钠($NaHCO_3$)以维持 80-120ppm 的碱度，初始投加量为每 1000 升水 135-200 克，并将泳池 pH 值控制在 7.4-7.6 之间。

Using a 1000L water tank to simulate a swimming pool cycle, pure salt and pure water are added to the equipment in the initial state, and electrolysis is carried out with a dual channel 15A current. To improve the hardness of the swimming pool water, 162g anhydrous calcium chloride (AR96%) and 60g anhydrous magnesium chloride (AR96%) are added every 1000L of water to simulate the "extremely hard water" in the United States (200mg/L calculated as CaCO3). According to three gradient tests, the first 50 hours of the first gradient (200mg/L calculated as CaCO3), the middle 50 hours of the second gradient (400mg/L calculated as CaCO3), and the last 50 hours of the third gradient (600mg/L calculated as CaCO3) are added in one go. The dosing method is

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based on monitoring data. (Handheld hardness tester) supplement method, record hardness readings every two hours; Add sodium bicarbonate (NaHCO3) to maintain an alkalinity of 80-120ppm, with an initial dosage of 135-200 grams per 1000 liters of water, and control the pH value of the swimming pool between 7.4-7.6.

10/13 SWC：每次换水后（0、25、50、75、100、125 小时）：先加碳酸氢钠使总碱度达到 80-120 mg/L、pH 7.4-7.6，再按方法 4 测余氯，然后按该阶段目标补足硬度，再开机。每 1000 L 投加量：200 mg/L 为 162 g 氯化钙+60 g 氯化镁；400 mg/L 为 324 g+120 g；600 mg/L 为 486 g+180 g；碳酸氢钠 135-200 g。硬度用百灵达钙硬度和总硬度药片（Calcicol、Hardicol）或 EDTA 滴定盒检测，超过 500 mg/L 时样品用蒸馏水 1:1 稀释。五合一水质仪不能测硬度。

10/13 SWC: After each water change (0, 25, 50, 75, 100 and 125 h), dose NaHCO3 to 80-120 mg/L alkalinity and pH 7.4-7.6, run the chlorine check in method 4, then dose hardness to the phase target before restarting. Per 1000 L: 162 g CaCl2 + 60 g MgCl2 for 200 mg/L, 324 g + 120 g for 400 mg/L, 486 g + 180 g for 600 mg/L, and 135-200 g NaHCO3. Measure hardness with Palintest Calcicol and Hardicol tablets or an EDTA titration kit. Above 500 mg/L, dilute the sample 1:1 with distilled water. The five-in-one meter doesn't read hardness.

10/13 SWC：维持泳池 pH 和碱度。满负荷（2 x 15 A）时，过氧化氢脱氯后每小时最多约 1.1 mol 酸进入水箱，消耗约 56 mg/L 碱度，初始碱度 2-3 小时即用完。pH 下降后碳酸钙垢无法形成，硬度梯度测试就失去意义，因此需要持续补碱。优先用第二台蠕动泵随电极连续投加碱液（例如用蒸馏水配制的 150 g/L 碳酸钠溶液，约 4-7 mL/min），根据每 2 小时的 pH 和碱度读数调节泵速；如不可行，每 2 小时人工补加碳酸氢钠。不排碱时全程最多约需 14 kg 碳酸氢钠（或 9 kg 碳酸钠），方法 3 的定时排碱会减少用量。

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记录每次投加的时间、质量或体积和浓度。

10/13 SWC: Holding pool pH and alkalinity. At full output (2 x 15 A), once H2O2 has removed the chlorine, up to about 1.1 mol of acid per hour reaches the tank. That uses about 56 mg/L of alkalinity an hour, so the starting alkalinity is gone in 2-3 h. Once pH drops, CaCO3 scale can't form and the hardness steps stop testing what they are meant to, so base has to go in all the time. The preferred way is a second peristaltic pump that runs with the electrodes, dosing for example a 150 g/L soda ash solution made with distilled water at about 4-7 mL/min, trimmed from the 2 h pH and alkalinity readings. If that isn't practical, add NaHCO3 by hand at each 2 h check. With no caustic drain the run needs up to about 14 kg of NaHCO3 (or 9 kg of soda ash), and the scheduled drain in method 3 should cut this. Log every addition with time, mass or volume, and concentration.







备注 Note:：

a、关于硬度，以碳酸钙（$CaCO_3$）计，单位为 mg/L；我了解到以下信息（如果有错误请告知我修改）：

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美国地质调查局（USGS）将硬度分为四档：软水：0-60 mg/L，中硬度水：60-120 mg/L，硬水：120-180 mg/L，极硬水：> 180 mg/L；水中钙镁离子比例为 70%钙 + 30%镁，我们在实验中模拟 200mg/L 碳酸钙含量的泳池水属于极硬水~~，这将加速老化~~ [10/13 SWC]；

Regarding hardness, it is measured in mg/L using calcium carbonate (CaCO3); I have learned the following information (please let me know if there are any errors):
The United States Geological Survey (USGS) classifies hardness into four levels: soft water: 0-60 mg/L, medium hardness water: 60-120 mg/L, hard water: 120-180 mg/L, and extremely hard water:>180 mg/L; The proportion of calcium and magnesium ions in the water is 70% calcium+30% magnesium. In our experiment, we simulated that pool water with a calcium carbonate content of 200mg/L belongs to extremely hard water~~, which will accelerate aging~~ [10/13 SWC];

10/13 SWC：USGS 分档针对自来水，不针对泳池水。泳池水因蒸发浓缩和反复补水会更硬，自来水硬的地区更明显。600 mg/L 偏高但真实存在，因此本测试按最恶劣工况测试，而不是加速老化。

10/13 SWC: The USGS bands describe tap water, not pool water. Pool water runs harder, because evaporation concentrates the minerals and every refill adds more, faster where the tap water is hard. 600 mg/L is high but real, so this test treats it as worst-case testing, not accelerated aging.

b、在 1000L 模拟泳池中，调配到 200mg/L 碳酸钙浓度的极硬水；需要投加钙离子（Ca^{2+}）需求：200g × 70% × (40/100) = 56.0 克；镁离子（Mg^{2+}）需求：200g × 30% × (24.3/100) = 14.6 克；因此需要投加 162g 无水氯化钙（AR96%）和 60g 无水氯化镁

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（AR96％）；

In a 1000L simulated swimming pool, the extremely hard water with a concentration of 200mg/L calcium carbonate is mixed; Need to add calcium ions (Ca^{2+}): 200g x 70% x (40/100)=56.0g; Magnesium ion (Mg^{2+}) demand: 200g × 30% × (24.3/100)=14.6 grams; Therefore, 162g anhydrous calcium chloride (AR96%) and 60g magnesium chloride (AR96%) need to be added;

2、测试中使用一台 DPD 样机（东润/博取）10 分钟测量间隔，连接泳池循环管路，进行余氯监测，直接将测试数据实时发送到 DaMonTo，并且每两小时手动记录一次 DPD 机器数据；同时，记录每次脱氯剂的添加时间、体积和浓度以便我们可以计算我们产生的氯需求。

During the test, a DPD prototype (~~Dongrun/Bochu~~BOQU [10/13 SWC]) was used with a 10 minute measurement interval. The pool circulation pipeline was connected for residual chlorine monitoring, and the test data was directly sent in real-time to DaMonTo. DPD machine data was manually recorded every two hours; At the same time, record the addition time, volume, and concentration of each dechlorination agent so that we can calculate the chlorine demand we generate.

10/13 SWC：现场分析仪为博取（BOQU），清单中的试剂为 SWCL-300，请确认二者匹配。10 分钟间隔 150 小时约 900 次测试，加冲洗和校准约 1000 次，请确认试剂量足够。将分析仪的设备 ID 发给 Incodeks。

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10/13 SWC: The analyzer on site is a BOQU and the listed reagents are for the SWCL-300, so please confirm they match. A 10-minute interval over 150 h is about 900 tests, about 1,000 with flushes and calibration. Please confirm the reagent set covers that. Send the analyzer's device ID to Incodeks.

3、开始在第一梯度 50 小时（硬度浓度 200mg/L）和第二梯度 50 小时（硬度浓度 400mg/L）和第三梯度 50 小时测试（硬度浓度 600mg/L）测试中使用正常软件(V 1.10.132)，这个版本的软件包含结束运行后冲洗 5 分钟。

关于排碱测试，在主测试之后(测试完三阶段 150 小时后)，再分阶段测试 pH 触发式排放装置和~~腐蚀性水平传感器~~碱液液位传感器 [10/13 SWC]。同步安排在 HASA 池或 Urim 的池中进行这些测试。至于判断 PH 排放的阈值（请注意，在 7/2 清洁测试中，盐水 pH 值约为 10.3），根据主测试流程记录的 PH 值数据再定；

Start using normal software (V 1.10.132) in the first 50 hour gradient test (hardness concentration 200mg/L), the second 50 hour gradient test (hardness concentration 400mg/L), and the third 50 hour gradient test (hardness concentration 600mg/L). This version of the software includes a 5-minute rinse after running.
Regarding the alkali discharge test, after the main test (150 hours after completing the three-stage test), the pH triggered discharge device and ~~corrosive level sensor~~caustic level sensor [10/13 SWC] will be tested in stages. Synchronously schedule these tests in the HASA pool or Urim pool. As for the threshold for determining pH emissions (please note that in the 7/2 cleaning test, the pH value of saline water is about 10.3), it will be determined based on the pH data recorded in the main testing process

10/13 SWC：排碱。我们读到的主板固件（至 v1.9.108）不会自动打开 D 阀，只能在屏幕的 I/O 测试页手动打开，与 HSW 10/8 的

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判断一致。请 Incodeks 确认 v1.10.132 是否相同，以及运行结束后的冲洗包括哪些动作。主测试中请定时手动排碱：间隔和每次时长由 HSW 根据碱液箱容积提出；排放时增压泵运行，使碱液按设计经文丘里回到水箱。每次记录时间、排放量以及排放前后的盐水 pH。这样既控制盐水 pH，又把碱送回水箱中和酸。

10/13 SWC: Caustic drain. The mainboard firmware we can read, up to v1.9.108, never opens valve D on its own. It can only be opened by hand from the screen's I/O test page, which matches HSW's 10/8 note. Incodeks, please confirm the same holds for v1.10.132, and what the end-of-run rinse does. In the main test, please drain valve D by hand on a schedule. HSW to propose the interval and duration from the caustic tank volume. Keep the booster pump running during the drain so the caustic returns to the tank through the venturi, as designed. Log each drain with time, volume, and brine pH before and after. This keeps brine pH in check and sends the caustic back to neutralize the acid in the tank.

4、关于电解电极性能测试，考虑到需要先将水箱水更换成干净的水才能统计氯的产量，因此计划每运行 0 小时、25 小时、50 小时、75 小时、100 小时、125 小时、150 小时后进行换水，统计在两个电极都以 15A 电流工作，运行 10 分钟后，通过百灵达测试 1000L 泳池的余氯 FCL 读数(测试三次取平均数)，当测试出来的余氯读数低于 3.4mg/L 终止测试分析原因，并对电极进行拆解拍照记录；

Regarding the performance testing of electrolytic electrodes, considering the need to replace the water in the tank with clean water before calculating the chlorine production, it is planned to replace the water every 0 hours, 25 hours, 50 hours, 75 hours, 100 hours, 125 hours, and 150 hours of operation. The FCL reading of residual chlorine in a 1000L swimming pool will be measured by testing with a current of 15A for both electrodes after 10 minutes of operation (taking the average of three tests). When the residual chlorine reading is below 3.4mg/L, the test will be terminated to analyze the cause, and the electrodes will be disassembled and photographed for recording;

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10/13 SWC：余氯测试的固定条件：(1) 新水补碱后 pH 7.4-7.6（见方法 1），记录水温和盐水温度；(2) 测试期间关闭脱氯剂泵和碱液泵，水中不能残留过氧化氢；(3) 在远离回水口和投加口的固定取样点，开机前和运行 10 分钟后各测一次，结果取升幅；(4) 每次取样测三次取平均，超出量程时用蒸馏水 1:1 稀释。参考：理论升幅约 6.6 mg/L，产品指标 4.4 mg/L。低于 4.4 mg/L 标记并报告，低于 3.4 mg/L 停止测试。

10/13 SWC: Fixed conditions for the chlorine check. (1) Fresh water, dosed to pH 7.4-7.6 as in method 1, with water and brine temperatures recorded. (2) Dechlorinator and base pumps off for the check, and no H2O2 left in the water. (3) One fixed sample point away from the return and the dosing points, read before the run and after 10 minutes. The result is the rise. (4) Three readings averaged each time. Above the photometer range, dilute 1:1 with distilled water. For reference, the theoretical rise is about 6.6 mg/L and the product spec is 4.4 mg/L. Below 4.4 mg/L, flag and report it. Below 3.4 mg/L, stop the test.

5、关于脱氯我们已经在国内寻找硫代硫酸钠，如下图我们可以购买 AR 级纯度，但关于投加方式需要进一步给您讨论，或者您是否有好的建议，目前现场条件有限我们可能会采用人工一次性投加后期补加的方式，使泳池余氯维持 2-4PPM 左右，请您审核后告诉我们您的意见；

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10/8 补充：使用变速蠕动泵持续投加脱氯剂(尝试使用过氧化氢，使用过程有问题再调整为硫代硫酸盐)。通过电极输出端的固态继电器(SSR)为泵供电，这样当电极产生氯时泵就会运行。用蒸馏水将脱氯器配制到已知浓度，并根据 DPD 读数手动调节泵速，以保持泳池 FCL 稳定。通过电极设置、投加记录和 DPD 数据，即使泳池 FCL 没有变化，我们也能准确计算氯的产量。选型蠕动泵和装脱氯剂的桶需要先知道每分钟需要添加多少过氧化氢（我们可以采购到 4%，6.5%，7.5%三种规格的双氧水），您对蠕动泵的规格和脱氯剂的桶有什么建议吗？如果采购额外的定时继电器，以便可以循环定时添加，而不是一直在添加，这种方案是否可行呢？

We have been searching for sodium thiosulfate for dechlorination in China. As shown in the figure below, we can purchase AR grade purity. However, we need to further discuss the addition method with you, or do you have any good suggestions? Currently, due to limited on-site conditions, we may use manual one-time addition and later replenishment to maintain the residual chlorine in the swimming pool at around 2-4PPM. Please review and let us know your opinion;

10/8 Addendum: Use a variable speed peristaltic pump to continuously add dechlorination agent (try using hydrogen peroxide, adjust to thiosulfate if there are problems during use). The pump is powered by a solid-state relay (SSR) at the electrode output end, so that the pump will run when chlorine is generated by the electrode. Prepare the dechlorination device with distilled water to a known concentration and manually adjust the pump speed based on the DPD reading to maintain stable FCL in the swimming pool. Through electrode settings, dosing records, and DPD data, we can accurately calculate chlorine production even if the FCL of the swimming pool remains unchanged. Do you have any suggestions for the specifications of the peristaltic pump and the chlorine removal agent tank? We need to know how much hydrogen peroxide needs to be added per minute when selecting the peristaltic pump and chlorine removal agent tank (we can purchase hydrogen peroxide in three specifications: 4%, 6.5%, and 7.5%)? Is it feasible to purchase additional timing relays so that they can be added cyclically instead of continuously?

10/13 SWC：过氧化氢投加量。7.5%过氧化氢在设计产氯量下约 2.8 mL/min，理论满负荷约 4.1 mL/min（1 mol H2O2 对应 1

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mol Cl2）。请选用约 1-10 mL/min 的变速蠕动泵，管路耐过氧化氢，由电极输出端的 SSR 控制，连续投加，不用定时继电器。

WS-20-03-M（20 L/h）只能用到约 1%额定流量，定时间歇投加会使 DPD 余氯读数波动。150 小时约需 26-38 kg 7.5%过氧化氢，

清单中 4 桶（10 kg）可用 40-60 小时，可分批采购。25 L 桶可用 4-6 天；桶盖要通气（过氧化氢会慢慢分解放出氧气），避免日晒。

如稀释，用蒸馏水配制，至少每 3 天配一次新液。

10/13 SWC: H2O2 dose. For 7.5% H2O2, plan on about 2.8 mL/min at design output and 4.1 mL/min at full theoretical output (1 mol H2O2 per mol Cl2). Use a variable-speed peristaltic pump of about 1-10 mL/min with tubing rated for H2O2, switched by the SSR from the electrode output and running continuously. No timer relay. The WS-20-03-M (20 L/h) would run at about 1% of its rating, and timed cycling makes the DPD chlorine readings swing. The full 150 h needs 26-38 kg of 7.5%. The four drums on the list (10 kg) cover 40-60 h, so buy in lots. A 25 L drum lasts 4-6 days. Vent the drum lid, because H2O2 slowly releases oxygen, and keep it out of the sun. If you dilute it, use distilled water and mix a fresh batch at least every three days.





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6、使用在线温度检测仪器对增压水泵，PCBA，进行不间断温度监测将含匹配时间戳的读数保存到电脑当中以备后续查看，同时分时段使用手持温度传感器对盐水温度进行检测并记录（盐水腐蚀性可能无法进行在线连续监测）；

10/8 补充：匹配时间戳是关键部分，因此在开始时将记录仪时钟设置为 DaMonTo 的时间，之后我们可以对数据进行对齐。对于盐水，采购专为恶劣环境设计的涂层温度传感器。

Using online temperature detection instruments for the booster water pump, PCBA, Perform uninterrupted temperature monitoring and save the readings with matching timestamps to the computer for future viewing. At the same time, use handheld temperature sensors to detect and record the temperature of saltwater in different time periods (saltwater corrosiveness may not be continuously monitored online);10/8 Addendum: Matching timestamps is a crucial part, so set the recorder clock to the time of DaMonTo at the beginning, and then we can align the data. For saltwater, purchase coated temperature sensors specifically designed for harsh environments.

10/13 SWC：在现有多参数温度监测仪上连续记录：增压水泵、PCBA、两路电源、控制箱、水箱水温和环境温度，另加环境湿度。盐水温度探头选用 PTFE 涂层或钛护套、适用于含氯饱和盐水的探头，装在盐水箱靠近电解槽一侧；清单中的“铝水锌水专用”探头不适用。所有记录仪时钟统一设为 UTC（与 DaMonTo 一致）；人工记录可用北京时间，但需注明 UTC+8。

10/13 SWC: Log continuously on the multi-channel temperature monitor already on site: booster pump, PCBA, both power

supplies, control box, tank water and ambient, plus ambient humidity. For brine, use a PTFE-coated or titanium-sheathed probe rated for saturated brine with dissolved chlorine, placed on the cell side of the brine tank. The probe on the list is made for molten aluminium and zinc and won't do. Set every logger clock to UTC, matching DaMonTo. Manual sheets can stay in Beijing time if they are marked UTC+8.

7、在长时间运行中关于监测数据阀值的设定，开始之前先记录所有数据的初始值， 然后再开始测试和记录这些数据。当氯的产量(低于 3.4mg/L),流量计 A(衰减百分之 20），压力传感器（衰减百分之 20），各部件监测温度（超过 70 摄氏度），水样变色（肉眼可见），电解电压（增加超过百分之 40），~~水 pH 值(大于 11)~~盐水 pH 值 PH1(大于 11) [10/13 SWC]以上到达任一监测阀值终止测试，分析原因谈论后决定是否需要更换部件继续测试；

Regarding the setting of monitoring data thresholds during long-term operation, record the initial values of all data before starting testing and recording these data. When the chlorine production (less than 3.4mg/L), flow meter A (attenuation of 20%), pressure sensor (attenuation of 20%), temperature monitoring of various components (exceeding 70 degrees Celsius), water sample discoloration (visible to the naked eye), electrolysis voltage (increased by more than 40%), and ~~water pH value (greater than 11)~~brine pH, PH1 (greater than 11) [10/13 SWC] reach any monitoring threshold, the test is terminated. After analyzing the reasons and discussing, it is decided whether to replace the components to continue the test;

10/13 SWC：余氯测试低于 4.4 mg/L 时先标记并报告，低于 3.4 mg/L 停止。每次 2 小时检查时，用白卡作背景拍摄水样，作为水样变色的判断依据。

10/13 SWC: A chlorine check below 4.4 mg/L is flagged and reported, and below 3.4 mg/L the test stops. At each 2 h

check, photograph a water sample against a white card, so discoloration is judged against a reference.

8、设备老化过程将模拟用户真实使用环境，氯的产出和氢气的产出都汇总到文丘里管 T 型口出，通过增压水泵驱动统一排放，整个老化过程需要人员值守，~~当发现异常情况需终止测试，~~ [10/13 SWC]异常行为应触发暂停和检查，然后由工程团队共同决定:修复、按现状继续，或判定为故障并结束测试。如果更换了部件，将更换后的时间作为独立段记录，并保留旧部件以供拆解分析。购买氯气和氢气检测和报警装置，用于记录外壳内的 Cl2 和 H2 数据，~~其中 Cl2 传感器显示低值，H2 传感器在泄压阀附近显示高值~~ Cl2 传感器装在低处，H2 传感器装在泄压阀附近高处 [10/13 SWC]。这是我们的内部泄漏检测，对现场安全和腐蚀问题至关重要。

The aging process of the equipment will simulate the real usage environment of the user. The production of chlorine and hydrogen will be collected at the T-shaped outlet of the Venturi tube and discharged uniformly through the booster water pump. The entire aging process requires personnel to be on duty. ~~When abnormal situations are found, the test should be terminated,~~ [10/13 SWC]and abnormal behavior should trigger a pause and inspection. Then, the engineering team will jointly decide whether to repair, continue as is, or determine as a fault and end the test. If a component is replaced, record the time after replacement as a separate segment and keep the old component for disassembly and analysis. Purchase chlorine and hydrogen detection and alarm devices for recording Cl2 and H2 data inside the casing, ~~where the Cl2 sensor displays a low value and the H2 sensor displays a high value near the pressure relief valve~~with the Cl2 sensor mounted low and the H2 sensor mounted high near the pressure relief valve [10/13 SWC]. This is our internal leak detection, which is crucial for on-site safety and corrosion issues.

10/13 SWC：探测器需要有输出（4-20 mA、RS-485 或内置记录），使 Cl2 和 H2 读数可以连续记录；清单中的报警器只能显示和报警。建议用报警继电器就地切断电解电源，不依赖 Wi-Fi 或后台。现场安全由 HSW 负责。

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10/13 SWC: The detectors need an output (4-20 mA, RS-485 or a built-in logger) so the Cl2 and H2 readings are logged. The units on the list only display and alarm. We recommend wiring the alarm relay to cut the electrode supply locally, so it doesn't depend on Wi-Fi or the backend. Site safety stays with HSW.

9、根据现有的测试条件，我们药品投加和数据采集以人工为主，以操作的工作量来计算，采集的周期不能低于两个小时一次，主要采集设备是 DPD 在线余氯监测仪，手持 pH 计， 百灵达余氯检测仪，手持温度传感器，手持电流感应监测器，在线温度监测器等；

The unit already sends flow, pressure, cell voltage and current, PH1, EC1, EC2, ORP and valve events to DaMonTo with 1minutes interval

According to the existing testing conditions, our drug dosing and data collection are mainly manual, calculated based on the workload of the operation. The collection cycle cannot be less than once every two hours. The main collection equipment includes DPD online residual chlorine monitoring device, handheld pH meter, Bailingda residual chlorine detector, handheld temperature sensor, handheld current sensing monitor, online temperature monitor, etc;

10/13 SWC：整机每 1 分钟向 DaMonTo 上报流量、压力、电解电压和电流、PH1、EC1、EC2、ORP 和阀门动作。DaMonTo 和记录仪的数据为主记录，人工读数用于校核。A、B 两槽请分开记录。请将 2 号机和 DPD 分析仪的设备 ID 以及上报间隔发给 Incodeks（Arlind），用于 HCG-381 数据归档；DaMonTo 原始消息只保存 90 天。

10/13 SWC: DaMonTo and the loggers are the record, and manual readings check them. Keep cells A and B as separate

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channels. Send the device IDs of Unit 2 and the DPD analyzer, and the report interval, to Incodeks (Arlind) for the HCG-381 archive. Raw DaMonTo messages are deleted after 90 days.

10、测试结束后对电解电极和文丘里管进行拆解，记录图片变化；

10/8 补充：测试结束后，检查并拍摄电极、膜、文丘里管、增压泵、阀门内部结构以及若干管道样品，以观察结垢或腐蚀情况。排空整个装置，测量剩余盐水及水箱中残留碱液的体积和 pH 值。然后检查空水箱是否存在结垢、杂物或损坏。

After the test is completed, disassemble the electrolytic electrode and Venturi tube, and record the changes in the pictures;
10/8 Addendum: After the test is completed, inspect and take photos of the electrodes, membranes, Venturi tubes, booster pumps, internal structures of valves, and several pipeline samples to observe scaling or corrosion. Drain the entire device and measure the volume and pH value of the remaining saltwater and alkaline solution in the water tank. Then check if there is scaling, debris, or damage in the empty water tank.

10/13 SWC：拆解时至少在盐水箱和文丘里 T 型口各取一份水垢样品，滴稀盐酸鉴别：碳酸钙会起泡，氢氧化镁溶解但不起泡。拆解盐水侧时佩戴防酸性气体面具。

10/13 SWC: At teardown, keep at least one scale sample each from the brine tank and the venturi T. A drop of dilute hydrochloric acid tells them apart: CaCO3 fizzes, Mg(OH)2 dissolves quietly. Wear a respirator with acid-gas cartridges when opening the brine side.

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记录参数 Record parameters：

- 流量计 A 数值，计划每 4 小时记录一次；

The value of flowmeter A is planned to be recorded every 4 hours;

- 压力传感器数值，计划每 4 小时记录一次；

The pressure sensor values are planned to be recorded every 4 hours;

- 电解电流及电压包含设备显示和实际测量，计划每 4 小时记录一次；

The electrolysis current and voltage include equipment display and actual measurement, and are planned to be recorded every 4 hours;

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- 盐水箱 PH1 数值（手持监测），计划每 4 小时记录一次；

PH1 value of salt water tank (handheld monitoring), planned to be recorded every 4 hours;

- 泳池水 PH2 数值，计划每 4 小时记录一次；

The PH2 value of the swimming pool water is planned to be recorded every 4 hours;

- ORP 传感器数值，计划每 4 小时记录一次；

ORP sensor values are planned to be recorded every 4 hours;

- 增压水泵温度（备注环境温度），计划每 4 小时记录一次；

The temperature of the booster pump (note the ambient temperature) is planned to be recorded every 4 hours;

- 盐水温度（手持温度计），计划每 4 小时监测一次；

Salt water temperature (handheld thermometer), planned to be monitored every 4 hours;

- EC1 传感器数值，计划每 8 小时记录一次；

EC1 sensor values are planned to be recorded every 8 hours;

- EC2 传感器数值，计划每 8 小时记录一次；

EC2 sensor values are planned to be recorded every 8 hours;

- DPD 余氯数值(测量间隔 10 分钟），计划每 2 个小时记录一次；

DPD residual chlorine value (measurement interval of 10 minutes), planned to be recorded every 2 hours;

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- 模拟泳池水电导率，计划每 2 小时记录一次；

Simulate the conductivity of swimming pool water, and plan to record them every 2 hours;

- 增压水泵温度，PCBA 温度，连续监测（备注环境温度），每 4 小时记录一次；

Boost water pump temperature, PCBA temperature, continuous monitoring (note ambient temperature), recorded every 4 hours;

- 称量我们添加的盐，并测量补充水量。单独使用 EC1 对于盐的使用来说不够可靠。

  weigh the salt you add and measure the makeup water volume. EC1 alone isn't reliable enough for salt use.

10/13 SWC：增加以下记录：泳池 pH 和总碱度，每 2 小时一次，同时记录每次补碱；硬度，每 2 小时一次（方法 1）；每次排碱的时间、排放量和前后盐水 pH（方法 3）；Cl2 和 H2 读数，连续记录（方法 8）；环境温度和湿度，连续记录；每次脱氯剂添加或泵速调整（方法 2、5）。

10/13 SWC: Also record pool pH and alkalinity every 2 h, with every base addition; hardness every 2 h (method 1); each caustic drain with time, volume and brine pH before and after (method 3); Cl2 and H2 continuously (method 8); ambient temperature and humidity continuously; and every dechlorinator addition or pump-speed change (methods 2 and 5).

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